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SRM Institute of Science and Technology
College of Engineering and Technology
School of Computing

Set - C

SRM Nagar, Kattankulathur – 603203, Chengalpattu District, Tamil Nadu
Academic Year: 2025-26 (EVEN SEM)

Test: LLT & FT- IV

Date: 29.04.2026

Course Code & Title:

Duration: 100 Minutes

Year & Sem: III Year / VI Sem

Max. Marks: 50

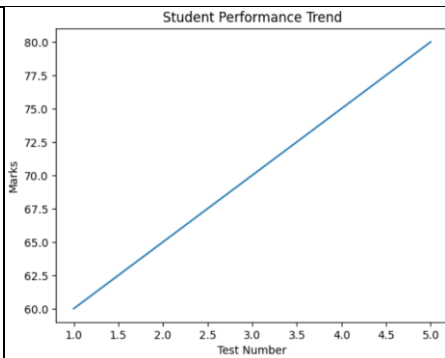
PART A—MCQ

Q.No Correct Option Answer

- | | | |
|----|---|--|
| 1 | C | Imputation + Duplicate removal + String manipulation |
| 2 | B | Z-score uses mean and standard deviation, Min-Max uses min and max |
| 3 | B | Melt → Pivot Table → Aggregation |
| 4 | B | IQR + Z-score methods |
| 5 | C | Chunk processing + filtering + aggregation |
| 6 | B | plt.text() |
| 7 | B | sns.kdeplot() |
| 8 | B | It shows the confidence interval around the median |
| 9 | B | plt.savefig("output.png", dpi=300) |
| 10 | C | To place a label for each data series on the plot |

Part – B (5 x 5 = 25 Marks)	
Instructions: Answer ANY 5 questions	
Q. No	Questions
11	A retail company maintains a dataset of customer names collected from different sources. The dataset contains inconsistencies such as uppercase and lowercase variations, leading/trailing spaces, and improper formatting. The analyst needs to standardize the text data. a) Define String Manipulation (1 Mark) b) Identify any two issues present in the dataset (1 Mark) c) Apply appropriate string operations to clean the data (2 Marks) d) Display the cleaned dataset (1 Mark) Answer: a) Definition (1 Mark) String manipulation is the process of modifying and cleaning text data using operations like trimming, case conversion, and formatting. b) Issues (1 Mark) • Uppercase/lowercase inconsistency • Leading/trailing spaces c) Cleaning Operations (2 Marks) <code>names = [" JOHN", "alice ", "BoB", " maRy "]</code> <code>cleaned = [name.strip().title() for name in names]</code> <code>print(cleaned)</code> d) Final Output (1 Mark) <code>['John', 'Alice', 'Bob', 'Mary']</code>
12	A school maintains a dataset of student marks. The management wants to analyze the overall performance by calculating summary statistics such as mean, median, and standard deviation. a) Create a sample dataset (1 Mark) b) Calculate summary statistics (2 Marks) c) Explain the importance of summarization in data analysis (1 Mark) d) Display the results (1 Mark) Answer: a) Dataset (1 Mark) <code>marks = [50, 60, 70, 80, 90]</code> b) Calculation (2 Marks) <code>import numpy as np</code> <code>mean = np.mean(marks)</code> <code>median = np.median(marks)</code> <code>std = np.std(marks)</code> <code>print(mean, median, std)</code> c) Importance (1 Mark) • Helps understand overall performance • Identifies central tendency and variation d) Final Output (1 Mark) <code>70.0 70.0 14.14</code>

13	A company maintains two datasets: one dataset contains employee details (Employee_ID, Name, Department) where Employee_ID is set as the index, and another dataset contains salary details (Employee_ID, Salary). The analyst needs to merge both datasets using the index. a) Create both datasets (1 Mark) b) Merge the datasets using index (2 Marks) c) Explain the concept of merging on index (1 Mark) d) Display the merged dataset (1 Mark) Answer: a) Create Datasets (1 Mark) import pandas as pd <pre>df1 = pd.DataFrame({ "Name": ["A", "B"], "Department": ["HR", "IT"] }, index=[1, 2])</pre> <pre>df2 = pd.DataFrame({ "Salary": [30000, 40000] }, index=[1, 2])</pre> b) Merge (2 Marks) merged = df1.join(df2) print(merged) c) Concept (1 Mark) Merging on index combines datasets using common index values d) Final Output (1 Mark) <pre> Name Department Salary 1 A HR 30000 2 B IT 40000 </pre>
14	A faculty member wants to monitor how student performance changes across multiple internal assessments conducted over a semester. The dataset contains marks scored by students in five consecutive tests, and the faculty needs a clear visualization to observe improvement or decline trends over time. Write a Python program to visualize the given data using a suitable plot. a) Create a dataset representing test numbers and corresponding marks (1 Mark) b) Plot an appropriate line graph to show performance trend (2 Marks) c) Add suitable axis labels and a meaningful title (1 Mark) d) Display the final output using appropriate function (1 Mark) Answer: a–d) Code (5 Marks) import matplotlib.pyplot as plt <pre> tests = [1, 2, 3, 4, 5] marks = [60, 65, 70, 75, 80] plt.plot(tests, marks) plt.xlabel("Test Number") plt.ylabel("Marks") plt.title("Student Performance Trend") plt.show() </pre>



15 A data visualization appears too basic and lacks visual appeal, reducing its effectiveness in presentations. The analyst needs to evaluate and enhance the visualization using appropriate styling techniques to improve clarity and impact.

- Create a basic plot using a sample dataset. (1 Mark)
- Apply suitable styling techniques (e.g., themes, colors, gridlines, labels) to enhance the visualization. (2 Marks)
- Analyze how the applied styling techniques improve readability, clarity, and overall presentation quality. (1 Mark)
- Interpret the enhanced visualization and explain its effectiveness compared to the basic plot. (1 Mark)

Answer:

a) Basic Plot (1 Mark)

```
import matplotlib.pyplot as plt
```

```
months = ['Jan', 'Feb', 'Mar', 'Apr', 'May']
```

```
sales = [10, 15, 7, 12, 20]
```

```
plt.plot(months, sales)
```

```
plt.show()
```

b) Enhanced Visualization (2 Marks)

```
import matplotlib.pyplot as plt
```

```
months = ['Jan', 'Feb', 'Mar', 'Apr', 'May']
```

```
sales = [10, 15, 7, 12, 20]
```

```
plt.style.use('ggplot')
```

```
plt.figure(figsize=(8,5))
```

```
plt.plot(months, sales, color='blue', marker='o', linewidth=2)
```

```
plt.title('Monthly Sales Trend')
```

```
plt.xlabel('Months')
```

```
plt.ylabel('Sales')
```

```
plt.grid(True)
```

```
plt.show()
```

c) Analysis of Improvements (1 Mark)

- Theme improves professional appearance
- Labels/title improve understanding
- Gridlines support value comparison
- Colors/markers highlight data points

d) Interpretation (1 Mark)

- Sales trend fluctuates with highest value in May
- Enhanced plot is clearer and more engaging than basic plot

16	A visualization created by an analyst needs to be stored as an image file for inclusion in reports and presentations. The analyst must evaluate different saving options and ensure the output is suitable for high-quality usage. a) Create an appropriate plot using a sample dataset. (1 Mark) b) Save the plot using a suitable function with appropriate parameters. (2 Marks) c) Analyse different file formats (e.g., PNG, JPEG, PDF) and justify the most suitable format for reporting purposes. (1 Mark) d) Interpret the saved output and discuss how format and resolution affect the quality of the visualization. (1 Mark) Answer: a) Create Plot (1 Mark) import matplotlib.pyplot as plt # Sample data months = ['Jan', 'Feb', 'Mar', 'Apr', 'May'] sales = [10, 15, 7, 12, 20] plt.plot(months, sales, marker='o') plt.title('Monthly Sales') plt.xlabel('Months') plt.ylabel('Sales') b) Save Plot (2 Marks) plt.savefig('sales_plot.png', dpi=300, bbox_inches='tight') plt.show() c) Analysis of File Formats (1 Mark) • PNG: Lossless, high quality → best for reports and presentations • JPEG: Lossy compression → reduces quality (not ideal for graphs with text) • PDF: Vector format → scalable without quality loss, ideal for publications Justification: PNG or PDF are most suitable; PNG preferred for slides, PDF for print/publications d) Interpretation (1 Mark) • Saved image clearly represents the sales trend • Higher DPI (300) ensures sharp text and lines • Low resolution → blurry visualization • Format choice impacts clarity: ○ JPEG → may distort text/lines ○ PNG/PDF → preserves accuracy and readability
17	A data analyst is working with a dataset containing multiple numerical features such as sales, profit, and customer count. The analyst wants to visualize relationships between variables while also comparing individual trends across different plots. However, the analyst is unsure whether to use multiple plots (subplots) or a pair plot for effective visualization and deeper analysis. Based on the scenario, compare multiple plots and pair plot with suitable examples. a) Define Multiple Plot (Subplots) (1 Mark) b) Define Pair Plot (1 Mark) c) Compare both techniques (any two points) (2 Marks) d) Suggest the most suitable technique for analyzing relationships among multiple variables with justification (1 Mark) Answer: a) Multiple Plot (1 Mark) • Displaying multiple graphs in one figure using subplots b) Pair Plot (1 Mark) • Displays relationships between all variables in dataset c) Comparison (2 Marks)

Multiple Plots	Pair Plot
Manual plotting	Automatic relationships
Limited comparison	Shows all variable relationships

d) Best Technique (1 Mark)
Pair Plot
Justification

- Shows **correlation between multiple variables**
- Ideal for **multi-feature analysis**

Part – C (1 x 15= 15 Marks) Instructions: Answer ALL the questions	
Q. No	Questions
18	A retail company maintains a large dataset of sales transactions with the following attributes: Order_ID, Product_Name, Product_Category, City, State, Sales_Amount, and Quantity. The management wants to analyze the data at a higher level to identify overall business trends. Instead of analyzing individual transactions, they require summarized information such as total sales by category, region-wise performance, and overall quantity sold. The analyst decides to apply data generalization and data aggregation to make the dataset more meaningful for decision-making. a) Define Data Generalization and Data Aggregation with suitable examples. (3 Marks) b) Explain the need and advantages of generalization and aggregation in large datasets. (4 Marks) c) Write a Python program to create a sample dataset and perform aggregation to calculate total and average sales by Product_Category or State. (4 Marks) d) Apply data generalization by converting detailed attributes (such as City to State or grouping categories) and explain the process. (2 Marks) e) Interpret the summarized output and explain key insights obtained from the aggregated data. (2 Marks) Answer: a) Definition with Examples (3 Marks) Data Generalization - Process of reducing detail level by converting data into higher-level concepts <i>Example:</i> City → State, Product → Category Data Aggregation - Process of summarizing data using operations like sum, average <i>Example:</i> Total sales by category b) Need & Advantages (4 Marks) Need - Large datasets are complex and time-consuming - Difficult to analyze individual transactions Advantages ✓ Reduces data size ✓ Improves analysis speed ✓ Helps identify overall trends ✓ Supports better decision-making c) Python Program (4 Marks) Code import pandas as pd

```
# Sample dataset
data = {
    "Product_Category": ["Electronics", "Clothing", "Electronics", "Clothing", "Furniture"],
    "State": ["TN", "TN", "KA", "KA", "TN"],
    "Sales_Amount": [1000, 500, 1500, 700, 800],
    "Quantity": [2, 3, 1, 4, 2]
}

df = pd.DataFrame(data)

# Aggregation
result = df.groupby("Product_Category").agg({
    "Sales_Amount": ["sum", "mean"],
    "Quantity": "sum"
})

print(result)
```

Final Output

	Sales_Amount		Quantity
	sum	mean	sum
Product_Category			
Clothing	1200	600.0	7
Electronics	2500	1250.0	3
Furniture	800	800.0	2

d) Data Generalization (2 Marks)

Example Code

```
# Adding City column
df["City"] = ["Chennai", "Chennai", "Bangalore", "Mysore", "Chennai"]

# Generalizing City → State
generalized = df.groupby("State")["Sales_Amount"].sum()

print(generalized)
```

Explanation

- Detailed data (**City**) is converted into higher level (**State**)
- Reduces complexity and gives **region-wise summary**

Output

```
KA    2200
TN    2300
```

e) Interpretation (2 Marks)

- **Electronics** has highest sales → most profitable category
- **Clothing** has higher quantity → high demand
- **TN state** has slightly higher sales than KA
- Helps in **business planning and resource allocation**

(OR)

19

A stock market analyst is analyzing historical stock price data collected over time. The analyst needs to visualize trends clearly by controlling axis limits, adjusting tick marks, adding annotations to highlight significant changes, and customizing line properties for better presentation. The final visualization should be clear, informative, and suitable for reporting.

- a) Explain the importance of axes, ticks, and labels in visualization (4 Marks)
- b) Write Python code to plot stock price trends (5 Marks)
- c) Add annotations and customize line properties (4 Marks)
- d) Interpret the visualization (2 Marks)

Answer:

a) Importance of Axes, Ticks, and Labels (4 Marks)

Axes

- Represent the **scale and dimensions** of data ($X \rightarrow \text{time}$, $Y \rightarrow \text{price}$)
- Help in understanding **data distribution**

Ticks

- Indicate **interval values** on axes
- Improve readability and help in **accurate interpretation**

Labels

- Provide **meaning to axes** (e.g., Time, Price)
- Make the graph **self-explanatory**

Overall Importance:

They make the visualization **clear, readable, and informative**

b) Python Code to Plot Stock Trend (5 Marks)

```
import matplotlib.pyplot as plt
```

```
# Sample data
```

```
days = [1, 2, 3, 4, 5]
```

```
prices = [100, 120, 115, 130, 150]
```

```
plt.plot(days, prices)
```

```
plt.xlabel("Days")
```

```
plt.ylabel("Stock Price")
```

```
plt.title("Stock Price Trend")
```

```
plt.show()
```



c) Add Annotation & Customize Line (4 Marks)

```
import matplotlib.pyplot as plt
```

```
days = [1, 2, 3, 4, 5]
```

```
prices = [100, 120, 115, 130, 150]
```

```
plt.plot(days, prices, linestyle='--', marker='o')
```

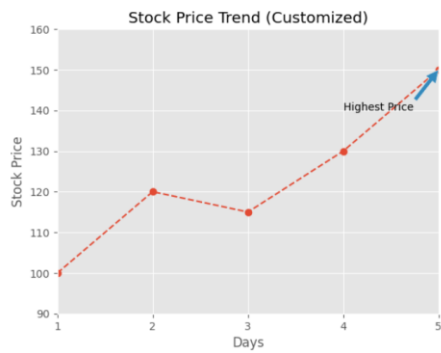
```
# Annotation
```

```
plt.annotate("Highest Price", xy=(5,150), xytext=(4,140), arrowprops=dict()))
```

```
# Axis limits & ticks
```

```
plt.xlim(1, 5)
```

```
plt.ylim(90, 160)
plt.xticks(days)
plt.xlabel("Days")
plt.ylabel("Stock Price")
plt.title("Stock Price Trend (Customized)")
plt.show()
```



d) Interpretation (2 Marks)

- Stock shows an **overall increasing trend**
- Slight dip at day 3 indicates **temporary decline**
- Highest price at day 5 shows **strong growth**